In [3]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline

df = pd.read_csv('vb_matches.csv')

print("Dataset Shape:", df.shape)
display(df.head())
print(df.info())
```

```
Dataset Shape: (76756, 65)
```

Columns 1–8 of 22

	circuit	tournament	country	year	date	gender	match_num
0	AVP	Huntington Beach	United States	2002	2002-05-24	M	1
1	AVP	Huntington Beach	United States	2002	2002-05-24	M	2
2	AVP	Huntington Beach	United States	2002	2002-05-24	M	3
3	AVP	Huntington Beach	United States	2002	2002-05-24	M	4
4	AVP	Huntington Beach	United States	2002	2002-05-24	M	5

Columns 9–15 of 22 · first column repeated

	w_player1	w_p1_birthdate	w_p1_age	...	l_p1_tot_blocks	l_p1_tot_digs	l_p2_tot_attacks
0	Kevin Wong	1972-09-12	29.694730	...	0.0	NaN	NaN
1	Brad Torsone	1975-01-14	27.356605	...	2.0	NaN	NaN
2	Eduardo Bacil	1971-03-11	31.203285	...	1.0	NaN	NaN
3	Brent Doble	1970-01-03	32.386037	...	2.0	NaN	NaN
4	Albert Hannemann	1970-05-04	32.054757	...	0.0	NaN	NaN

Columns 16–22 of 22 · first column repeated

	l_p2_tot_kills	l_p2_tot_errors	l_p2_tot_hitpct	l_p2_tot_aces	l_p2_tot_serve_errors	l_p2_tot_blocks	l_p2_tot_digs
0	NaN	NaN	NaN	0.0	NaN	1.0	NaN
1	NaN	NaN	NaN	0.0	NaN	0.0	NaN
2	NaN	NaN	NaN	0.0	NaN	0.0	NaN
3	NaN	NaN	NaN	2.0	NaN	0.0	NaN

	l_p2_tot_kills	l_p2_tot_errors	l_p2_tot_hitpct	l_p2_tot_aces	l_p2_tot_serve_errors	l_p2_tot_blocks	l_p2_tot_digs
4	NaN	NaN	NaN	0.0	NaN	1.0	NaN

5 rows × 65 columns

```
<class 'pandas.DataFrame'>
RangeIndex: 76756 entries, 0 to 76755
Data columns (total 65 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   circuit                76756 non-null  str
 1   tournament             76756 non-null  str
 2   country                76756 non-null  str
 3   year                   76756 non-null  int64
 4   date                   76756 non-null  str
 5   gender                 76756 non-null  str
 6   match_num              76756 non-null  int64
 7   w_player1              76756 non-null  str
 8   w_p1_birthdate         76373 non-null  str
 9   w_p1_age               76373 non-null  float64
 10  w_p1_hgt               72790 non-null  float64
 11  w_p1_country           76744 non-null  str
 12  w_player2              76756 non-null  str
 13  w_p2_birthdate         76348 non-null  str
 14  w_p2_age               76348 non-null  float64
 15  w_p2_hgt               72740 non-null  float64
 16  w_p2_country           76751 non-null  str
 17  w_rank                 76608 non-null  str
 18  l_player1              76756 non-null  str
 19  l_p1_birthdate         75697 non-null  str
 20  l_p1_age               75697 non-null  float64
 21  l_p1_hgt               69768 non-null  float64
 22  l_p1_country           76738 non-null  str
 23  l_player2              76756 non-null  str
 24  l_p2_birthdate         75797 non-null  str
 25  l_p2_age               75797 non-null  float64
 26  l_p2_hgt               69773 non-null  float64
 27  l_p2_country           76746 non-null  str
 28  l_rank                 75516 non-null  str
 29  score                  76734 non-null  str
 30  duration               74507 non-null  str
 31  bracket                76756 non-null  str
 32  round                  71817 non-null  str
 33  w_p1_tot_attacks       14578 non-null  float64
 34  w_p1_tot_kills         14578 non-null  float64
 35  w_p1_tot_errors        14343 non-null  float64
 36  w_p1_tot_hitpct        14571 non-null  float64
 37  w_p1_tot_aces          16196 non-null  float64
 38  w_p1_tot_serve_errors  14339 non-null  float64
 39  w_p1_tot_blocks        16196 non-null  float64
 40  w_p1_tot_digs          14578 non-null  float64
 41  w_p2_tot_attacks       14582 non-null  float64
```

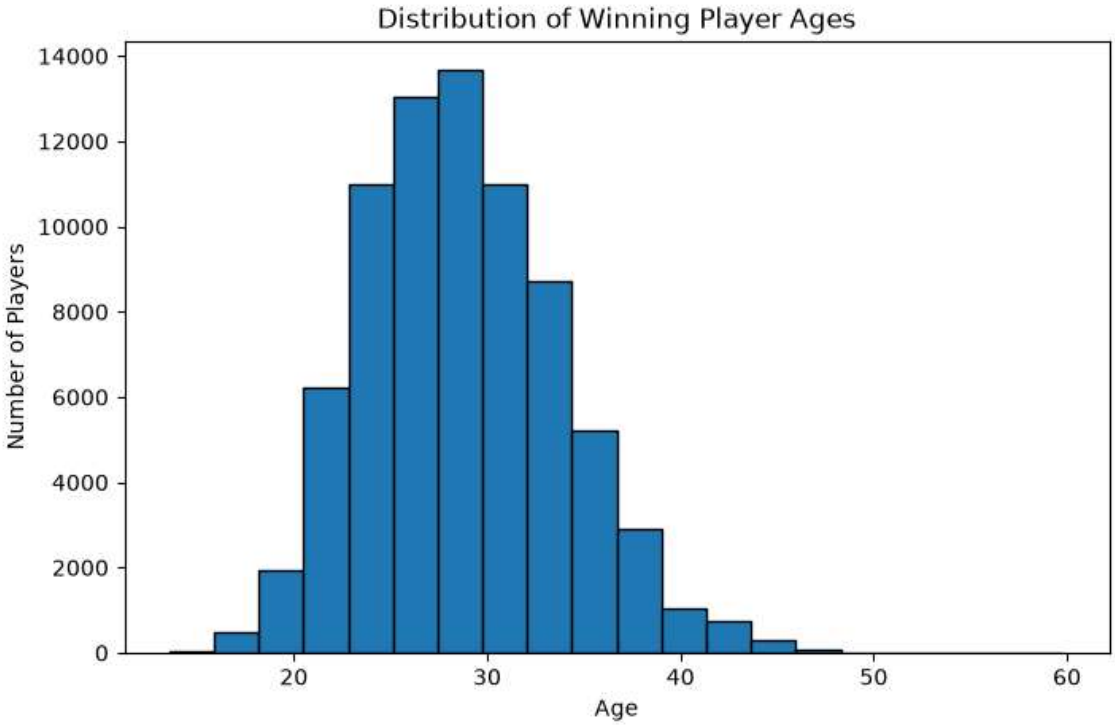
```
 42  w_p2_tot_kills         14582 non-null  float64
 43  w_p2_tot_errors        14343 non-null  float64
 44  w_p2_tot_hitpct        14575 non-null  float64
 45  w_p2_tot_aces          16200 non-null  float64
 46  w_p2_tot_serve_errors  14343 non-null  float64
 47  w_p2_tot_blocks        16200 non-null  float64
 48  w_p2_tot_digs          14582 non-null  float64
 49  l_p1_tot_attacks       14577 non-null  float64
 50  l_p1_tot_kills         14577 non-null  float64
 51  l_p1_tot_errors        14343 non-null  float64
 52  l_p1_tot_hitpct        14567 non-null  float64
 53  l_p1_tot_aces          16195 non-null  float64
 54  l_p1_tot_serve_errors  14338 non-null  float64
 55  l_p1_tot_blocks        16195 non-null  float64
 56  l_p1_tot_digs          14577 non-null  float64
 57  l_p2_tot_attacks       14578 non-null  float64
 58  l_p2_tot_kills         14578 non-null  float64
 59  l_p2_tot_errors        14343 non-null  float64
 60  l_p2_tot_hitpct        14567 non-null  float64
 61  l_p2_tot_aces          16196 non-null  float64
 62  l_p2_tot_serve_errors  14339 non-null  float64
 63  l_p2_tot_blocks        16196 non-null  float64
 64  l_p2_tot_digs          14578 non-null  float64
dtypes: float64(40), int64(2), str(23)
memory usage: 38.1 MB
None
```

In [4]:
```
plt.figure(figsize=(8, 5))

plt.hist(df['w_p1_age'].dropna(), bins=20, edgecolor='black')

plt.title('Distribution of Winning Player Ages')
plt.xlabel('Age')
plt.ylabel('Number of Players')

plt.show()
```

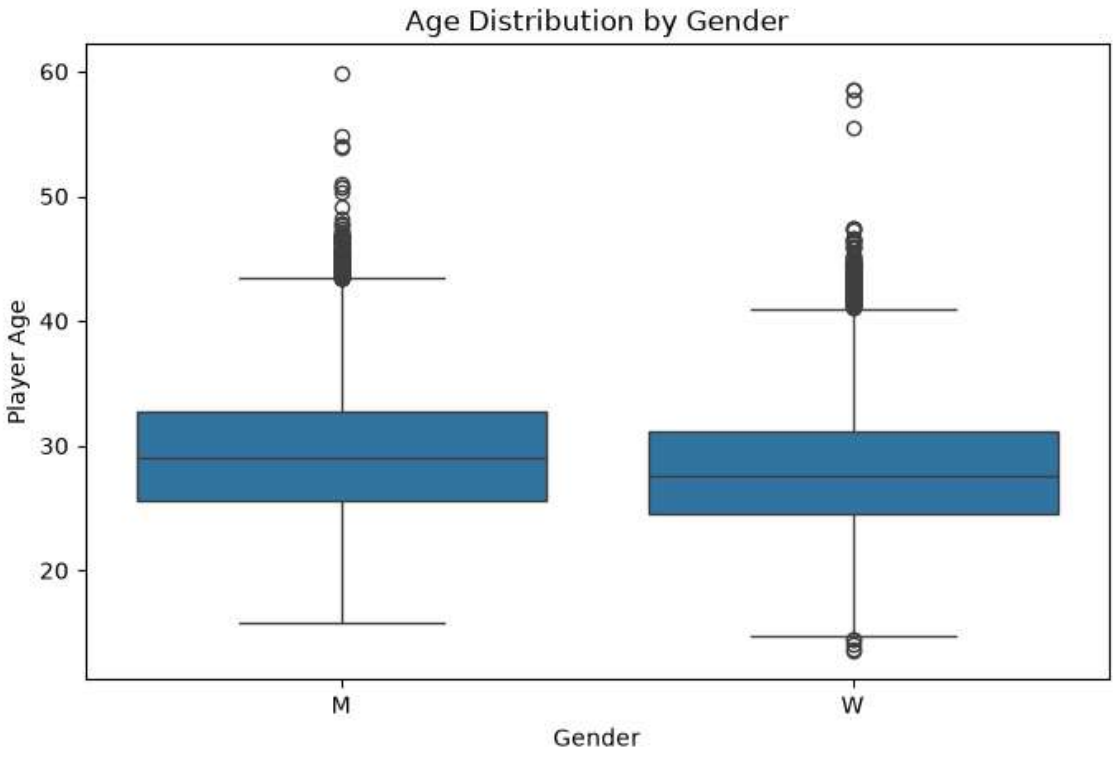


In [5]:
```
plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x='gender', y='w_p1_age')

plt.title('Age Distribution by Gender')
plt.xlabel('Gender')
plt.ylabel('Player Age')

plt.show()
```

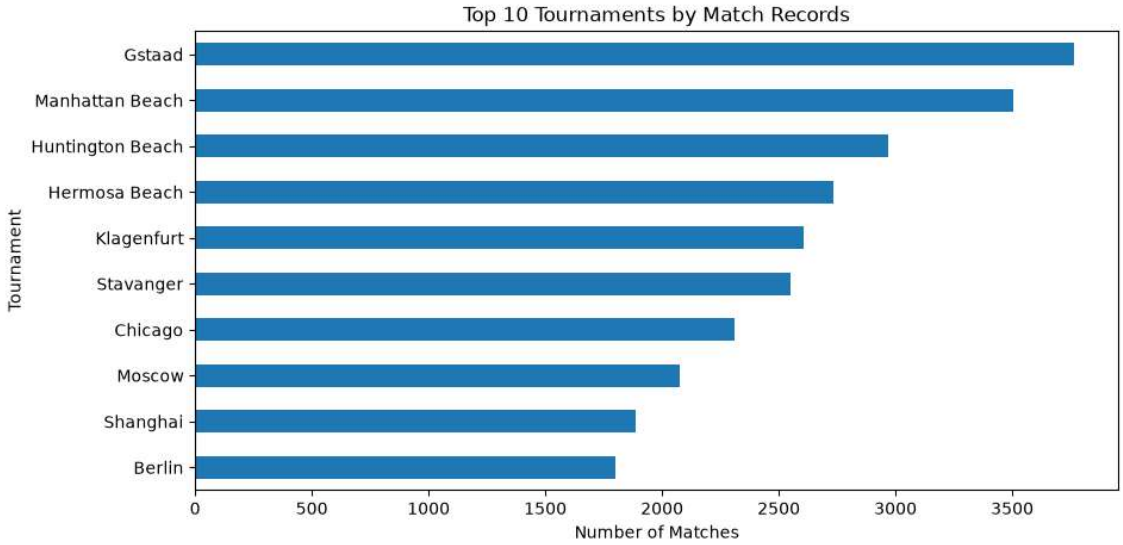


In [6]:
```
plt.figure(figsize=(10, 5))

df['tournament'].value_counts().head(10).sort_values().plot(
    kind='barh'
)

plt.title('Top 10 Tournaments by Match Records')
plt.xlabel('Number of Matches')
plt.ylabel('Tournament')

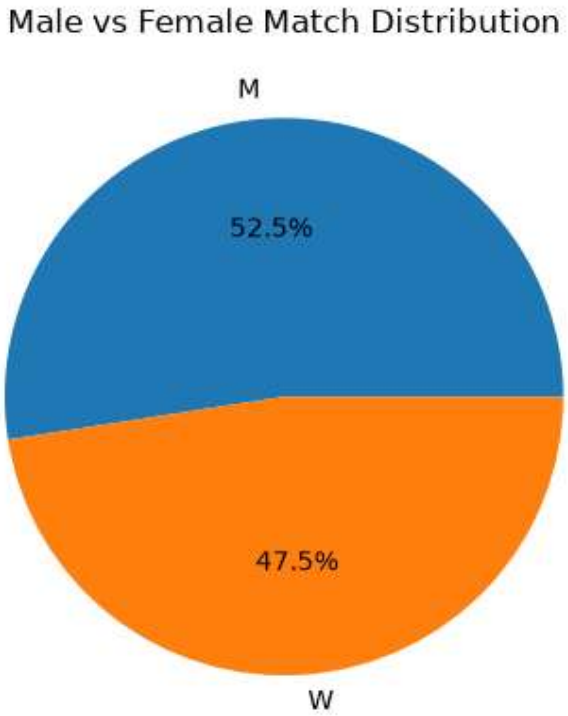
plt.show()
```



In [7]:
```
df['gender'].value_counts().plot(
    kind='pie', autopct='%1.1f%%'
)

plt.title('Male vs Female Match Distribution')
plt.ylabel('')

plt.show()
```

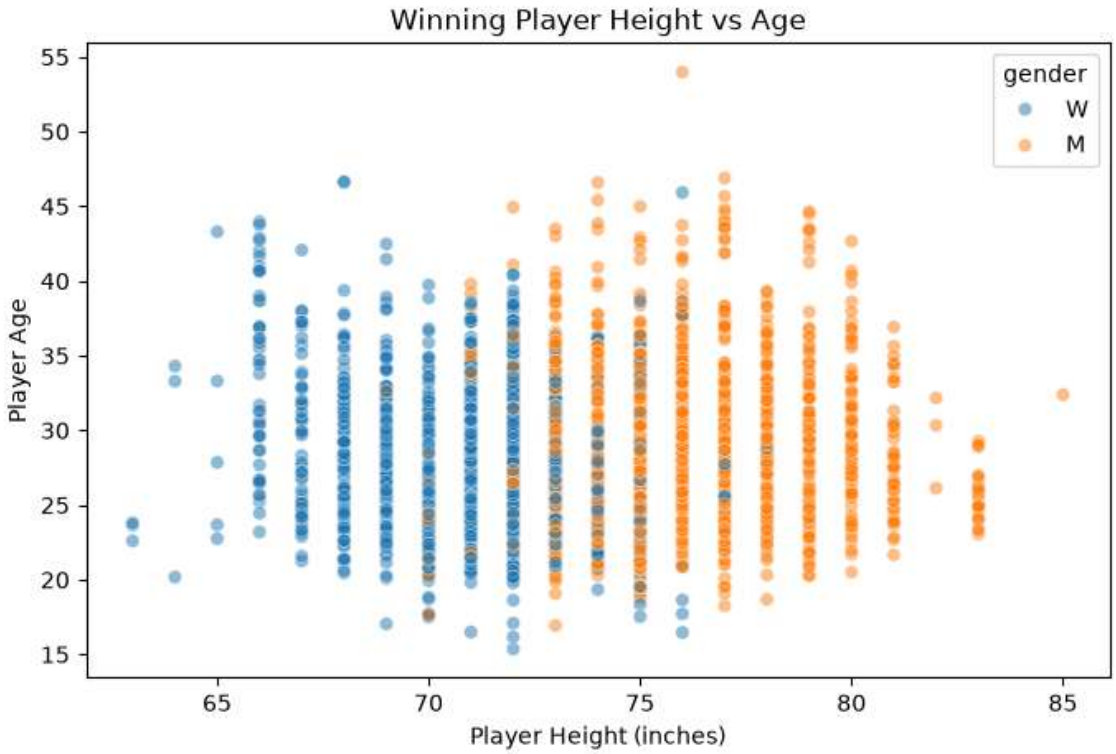


In [8]:
```
plt.figure(figsize=(8, 5))

sns.scatterplot(
    data=df.sample(n=min(3000, len(df)), random_state=42),
    x='w_p1_hgt',
    y='w_p1_age',
    hue='gender',
    alpha=0.5
)

plt.title('Winning Player Height vs Age')
plt.xlabel('Player Height (inches)')
plt.ylabel('Player Age')

plt.show()
```



In [9]:
```
matches = (
    df.groupby('year')
    .size()
    .sort_index()
)

plt.figure(figsize=(10, 5))

plt.plot(matches.index, matches.values, marker='o')

plt.title('Number of Volleyball Matches by Year')
plt.xlabel('Year')
plt.ylabel('Number of Matches')

plt.grid(True)

plt.show()
```

